THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The legs of a right triangle are 5 cm and 12 cm. Its hypotenuse is: 1

(a) 12 cm **(b)** 13 cm **(c)** 17 cm **(d)** 14 cm

2. The legs of a right triangle are 9 cm and 40 cm. Its hypotenuse is: 1

(a) 49 cm **(b)** 40 cm **(c)** 42 cm **(d)** 41 cm

3. The hypotenuse of a right triangle is 13 cm and one leg is 5 cm. The other leg is: 1

(a) 13 cm **(b)** 8 cm **(c)** 12 cm **(d)** 11 cm

4. The hypotenuse of a right triangle is 41 cm and one leg is 9 cm. The other leg is: 1

(a) 41 cm **(b)** 40 cm **(c)** 32 cm **(d)** 39 cm

5. Is a triangle with sides 7, 9 and 11 a right triangle? 1

(a) Yes **(b)** No

(c) Only if it is isosceles **(d)** Cannot say

6. The diagonal of a square of side 7 cm is: 1

(a) $7\sqrt{3}$ cm **(b)** 7 cm **(c)** 14 cm **(d)** $7\sqrt{2}$ cm

7. Baudhayana's Shulba Sutra describes how to: 1

(a) write numbers

(b) calculate profit

(c) construct altars using rectangles and squares

(d) measure time

8. Which is the longest side of a right triangle? 1

(a) median **(b)** base **(c)** altitude **(d)** hypotenuse

9. Multiplying a Pythagorean triplet by a constant gives: 1

(a) a square **(b)** another Pythagorean triplet

(c) a prime triplet **(d)** a non-triplet

10. Which of these is a Pythagorean triplet? 1

(a) (6, 7, 10) **(b)** (8, 15, 17) **(c)** (5, 7, 9) **(d)** (4, 6, 8)

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. A ladder 13 m long reaches 12 m up a wall. How far is its foot from the wall? 2

12. Find the diagonal of a square of side 6 cm. 2

13. How can you construct a square double the area of a given square? 2

14. What is the converse of the theorem? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. A ladder of length 25 m rests against a wall with its foot 15 m from the wall. How high up the wall does it reach? 3

16. A ladder of length 26 m rests against a wall with its foot 10 m from the wall. How high up the wall does it reach? 3

17. A rectangular park measures 3 m by 4 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

OR

A rectangular park measures 5 m by 12 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

18. A rectangular park measures 8 m by 15 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

OR

A rectangular park measures 7 m by 24 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) A rectangle has area 120 cm² and one side 15 cm. Find its diagonal.
(ii) Find its perimeter. 5

OR

Case study: A ship sails 24 km east and then 7 km north. 5
(i) How far is it from the start?
(ii) If it returns directly, how much distance is saved compared to going back the same way?
(iii) State the theorem used.

20. (i) Show that a square with side equal to the diagonal of a unit square has double area.
(ii) Draw/describe how to halve a square. 5

OR

(i) Prove that the triangle with sides 20, 99, 101 is right-angled. 5
(ii) Find its area.

END OF QUESTION PAPER Total: 40 Marks

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